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Quantitative Text Analysis (POL42050)

Level 4 Module; Spring Trimester 2025

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Latest version at: <https://muellerstefan.net/teaching/2025-spring-qta.pdf>

Time: Thu, 10:00–11:50

Location: [G107-ART \(Newman Building\)](#)

Credits: 10.0

Format: Lecture and computer labs

Module coordinator: Stefan Müller, PhD

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Office hours: Thur, 12:30–14:00 ([sign up here](#))

Course Content

Computational text analysis has become increasingly popular in political science in recent years. With the vast availability of text data on the web, political scientists increasingly view quantitative text analysis (or “text as data”) as a valuable approach for studying various forms of social and political behaviour.

This module introduces political science students to the quantitative analysis of textual data.¹ It covers the theoretical foundations, practical applications, technical implementations of these methods using the R statistical programming language, and how to validate text-as-data approaches. Most methods follow a three-step framework: first, identifying texts and units of analysis; second, extracting measurable features from these texts and converting them into a quantitative feature matrix; and third, analysing this matrix using statistical techniques, including dictionary-based approaches, scaling models, and topic modelling.

Building on this foundational framework, students will gain hands-on experience with advanced techniques such as word embeddings, transformer models, and generative AI. The module also incorporates the Hugging Face Python infrastructure, a leading resource for implementing transformer models and other state-of-the-art natural language processing tools.

Each session combines lectures with practical exercises, allowing students to apply these methods to political texts. These exercises address real-world challenges at each stage of the research process. By engaging with these approaches, students will gain insights into cutting-edge text analysis methods and their applications in political science research.

¹Note to researchers: You are welcome to reuse and build upon this document for your teaching. If you design a module based on this document, I would greatly appreciate it if you could reach out to me and include a link to this document in your syllabus.

Learning Outcomes

Upon successful completion of the course, students will be able to:

1. Understand fundamental issues in (quantitative) text analysis such as inter-coder agreement, reliability, validation, accuracy, and precision.
2. Convert texts into quantitative matrices of features, and then analyse those features using statistical methods, topic models, and scaling approaches.
3. Use human coding of texts to train supervised classifiers and fine-tune transformer models.
4. Apply these methods to their own text corpus to address a substantive research question.
5. Critically evaluate (social science) research that uses automated text analysis methods.

Prerequisites

Prior familiarity with the statistical programming language R (or Python) is a prerequisite for this course due to its direct relevance to the content and assignments. Below are some reasons why prior experience with R or Python is crucial for students to follow the course and apply the methods effectively:

- **Implementation of Text Analysis Methods:** Text analysis is a central component of the course, and R is widely used for implementing text analysis techniques. R provides a comprehensive set of libraries and packages specifically designed for text processing, natural language processing (NLP), and sentiment analysis. Students with prior experience in R will be able to navigate and utilize these tools more efficiently, enabling them to implement text analysis methods covered in the course effectively.
- **Course Content Alignment:** The course content, lectures, and materials are designed with a focus on R-based implementation. The examples, code snippets, and demonstrations provided throughout the course will be predominantly in R. Some of the advanced methods are implemented in Python, but a good understanding of R will make it much easier to write and run code in Python. Without prior familiarity, students may struggle to comprehend and replicate these examples, hindering their understanding of the core concepts and methodologies.
- **Homework Assignments and Research Papers:** The assignments and research papers in this course will require students to apply the text analysis methods discussed in class to real-world data. Students without prior experience with R may find it challenging to write code to preprocess large text corpora, visualise results, and interpret the findings. Their lack of proficiency in R could impede their ability to complete assignments accurately and efficiently.

General Readings

The seminar does not build on a single textbook, but relies on papers and book chapters. The following books and articles are recommended for a general overview of quantitative text analysis, natural language processing, and computational social science.

- K. L. Nielbo, F. Karsdorp, M. Wevers, A. Lassche, B. R. B., M. Kestemont, and N. Tahmasebi (2024). “[Quantitative Text Analysis](#)”. *Nature Reviews Methods Primers* 2 (24).
- K. Benoit (2020). “[Text as Data: An Overview](#)”. *Handbook of Research Methods in Political Science and International Relations*. Ed. by L. Curini and R. Franzese. Thousand Oaks: Sage: 461–497.

- K. Watanabe and S. Müller (2023). *Quanteda Tutorials*. URL: <https://tutorials.quanteda.io>.
- D. S. Stoltz and M. A. Taylor (2024). *Mapping Texts: Computational Text Analysis for the Social Sciences*. Oxford: Oxford University Press.
- E. Hvitfeldt and J. Silge (2021). *Supervised Machine Learning For Text Analysis in R*. Boca Raton: CRC Press.
- J. Grimmer, M. E. Roberts, and B. M. Stewart (2022). *Text as Data: A New Framework for Machine Learning and the Social Sciences*. Princeton: Princeton University Press.
- D. Jurafsky and J. H. Martin (2024). *Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition with Language Models*. 3rd edition.

Technical Background

The following books and websites are helpful to refresh and extend the knowledge of R, Quarto, and the `quanteda` package or move beyond the content covered in the module. Websites such as [Stack Overflow](#), [R bloggers](#), and the documentation of R packages will be helpful for solving practical problems. Most books listed in the syllabus are published in print, but also freely available online.

R, Quarto, and Data Science

- H. Wickham, M. Çetinkaya-Runde, and G. Grolemund (2023). *R for Data Science: Import, Tidy, Transform, Visualize, and Model Data*. 2nd edition. Sebastopol: O'Reilly.
- R. Alexander (2023). *Telling Stories with Data: With Applications in R*. New York: CRC Press.
- C. Ismay and A. Y. Kim (2020). *Statistical Inference via Data Science: A Modern Dive into R and the tidyverse*. Boca Raton: CRC Press.

Python for Social Scientists

- A. Turrell (2024). *Coding for Economists*. URL: <https://aeturrell.github.io/coding-for-economists>.

Data Visualisation

- K. Healy (2019). *Data Visualization: A Practical Introduction*. Princeton: Princeton University Press.

NLP with Transformers

- L. Tunstall, L. von Werra, and T. Wolf (2022). *Natural Language Processing with Transformers: Building Language Applications with Hugging Face*. Beijing: O'Reilly.

Software and Packages

In this module, we will run R and Python code in the integrated development environment (IDE) `VSCode`. We will also use `GitHub Copilot` (free to use for students).

On Brightspace, we have uploaded a detailed guidelines on installing the relevant software and extensions (see the **Installation Instructions** page under **My Learning**).

Important: Make sure to install and set up R, VSCode, and GitHub Copilot *before* our first lecture in Week 1. If you have any question related to (installing) the software, please ask this Question in our Slack workspace.

We will use the following packages in this module:

- Quantitative text analysis:
 - `quanteda`
 - `quanteda.textmodels`
 - `quanteda.textstats`
 - `quanteda.textplots`
- Importing and converting text files: `readtext`; `markdown`
- Topic models: `keyATM`; `BERTopic`
- Word embeddings: `wordvector`
- Data wrangling and visualisation: `tidyverse` (esp. `dplyr`, `tidyr`, `lubridate`, and `ggplot2`)
- Creating documents and reports: `Quarto`
- Applying and fine-tuning transformer-based machine learning models: `HuggingFace` & `transformers` (Python)

Plagiarism

Although this should be obvious, plagiarism – copying someone else’s text without acknowledgement or beyond ‘fair use’ quantities – is not allowed. Plagiarism is an issue we take very serious here in UCD. Please familiarize yourself with the definition of plagiarism on UCD’s website² and make sure not to engage in it.

Late Submission Policy

If a student submits an assignment late, the following penalties will be applied:

- Coursework received at any time within two weeks of the due date will be graded, but a penalty will apply.
 - Coursework submitted at any time up to one week after the due date will have the grade awarded reduced by two grade points (for example, from *B–* to *C*).
 - Coursework submitted more than one week but up to two weeks after the due date will have the grade reduced by four grade points (for example, from *B–* to *D+*). Where a student finds they have missed a deadline for submission, they should be advised that they may use the remainder of the week to improve their submission without additional penalty.
- Coursework received more than two weeks after the due date will not be accepted. Regulations regarding extenuating circumstances apply.

²<https://libguides.ucd.ie/academicintegrity>.

Office Hours

My office hours take place on Thursday from 12:30–14:00, either in person (Room G312, Newman Building) or online. Please sign up for a meeting at <https://calendly.com/mueller-ucd/office-hours>.

Questions and Problems

In this module, we will discuss concepts, methods, and software you might not have heard of before. I am aware that parts of this module could be challenging, and I will assist you as best as I can.

We will use Slack in this module.³ Make sure to create a Slack account before the first seminar and join the Slack workspace. If you have a question that involves code or concepts, please share your question in `#coding`, `#homework`, or `#research-paper`.

If you struggle to solve problems relating to R or RStudio, please follow the steps outlined below before contacting your peers or me. It is very likely that at least one other person faced the same problem before or received the same error message.

1. Try to summarise the problem in your own words and then google this summary. If the problem relates to R, add `rstats` to your search query. For example: `how to import csv file in rstats`. I am almost certain that you find a solution to most of your questions.
2. If your R code returns an error, I would advise you to Google the text of the error message. For example, you google the error message `"Error: Can't subset columns that don't exist."`

→ If steps 1–2 still do not solve your problem or question, please ask your question in the Slack channel devoted to this module. Your peers and I will help you.

Use of Generative Artificial Intelligence (AI) Tools

I encourage the use of generative AI tools when completing the assignments for this module but all work relying on AI-generated content must adhere to the highest academic standards. Users of this technology must be aware of what it can and more importantly what it cannot do well. It is crucial for you to exercise judgement when evaluating the quality and reliability of content generated through AI platforms. AI is not a panacea for all writing challenges; it will not automatically generate a flawless, logically coherent, and factually correct assignment. Instead, use AI as a tool to tackle specific issues such as brainstorming and idea formation, literature discovery, and text drafting issues. View your preferred AI platform(s) as useful but imperfect tools that can offer inspiration, new perspectives, and supplementary areas for research for your own work. In-depth research on your part remains essential to ensure coherent, factual, and scientifically informed perspectives in your assignment. Always cross-reference the information AI offers against other independent and reliable sources.

AI use must be in line with UCD's policies on academic integrity and adhere to the highest academic standards. See here for details: <https://libguides.ucd.ie/academicintegrity>.

Documenting AI Use (Mandatory)

Since generative AI is such a novel tool in an academic context, we do not yet fully understand what it is capable of, and these capabilities are evolving quickly. What was impossible today might well

³I have had very positive experiences with Slack in my modules. Müller (2023) discusses both the advantages and shortcomings of Slack for teaching and learning.

become trivial tomorrow (keeping in mind the academic standards mentioned above). In order to address this, you are expected to provide an account of the tools used and the way in which they were used in a mandatory appendix to your assignment. This appendix will be assessed as part of the assignment, with grade points awarded for effective communication of the methods used to generate content. For each instance where a generative AI tool is used, you need to provide:

1. An in-text citation or footnote. For example:
 - “Some AI-generated text (GPT-4o 2025)”
 - “Some AI-generated text⁴”
2. A bibliographic reference to the tool used and the date of access.
3. An entry in the mandatory AI appendix detailing how the tool was used. For example:

Table 1: Example table demonstrating how generative AI was used to complete the research paper

AI Tool	Explanation	Prompt used
ChatGPT	Topic brainstorming	“Provide an overview of potential political science research questions that could be answered using quantitative text analysis methods.”
Elicit	Literature search	“Compile a list of academic publications detailing advantages and shortcomings of topic models.”

Generative AI Tools

Below I have listed some AI tools that might help you drafting your research paper:

- Brainstorming
 - ChatGPT: <https://chat.openai.com>
 - Claude: <https://claude.ai>
- Literature Discovery
 - Elicit: <https://elicit.org>
 - Semantic Scholar: <https://www.semanticscholar.org>
 - Perplexity: <https://www.perplexity.ai>
 - SciSpace: <https://typeset.io>
- Structure and Drafting
 - Grammarly: <https://www.grammarly.com>
 - Quillbot: <https://quillbot.com/>
 - Jenni: <https://jenni.ai>

Syllabus Modification Rights

I reserve the right to reasonably alter the elements of the syllabus at any time by adjusting the reading list to keep pace with the course schedule. Moreover, I may change the content of specific sessions,

⁴Text based on content generated by OpenAI’s GPT-4o on 1 January 2025. See Appendix 1 for details.

depending on the participants' prior knowledge and research interests. If I make adjustments, I will email all seminar participants and upload the revised syllabus to Brightspace.

Dignity and Respect

UCD is committed to the promotion of an environment for work and study which upholds the dignity and respect of all members of the UCD community and which supports your right to study and/or work in an environment which is free of any form of bullying, harassment or sexual misconduct (including sexual harassment and sexual violence).

There are a number of supports in place if you are experiencing bullying, harassment or sexual misconduct and you are strongly encouraged to come forward to seek confidential support and guidance on the range of informal options and formal options for resolving issues as appropriate. Reports of bullying, harassment or sexual misconduct can also be made anonymously through UCD's Report and Support tool.

UCD is actively promoting a culture where bullying, harassment and sexual misconduct is not tolerated, where everyone is respected and feels valued, included and that they belong in UCD.

You can find more details on UCD's Dignity and Respect Website at: <https://www.ucd.ie/equality/support/dignityrespect/>.

Expectations and Grading

- Students are expected to read all papers or chapters assigned under **Mandatory Readings**. These readings serve as the basis for in-class discussions about the advantages, disadvantages, and applicability of the various approaches to social science questions. For each session, I also assign a variety of optional readings. I strongly encourage students to (at least) skim these readings. Both the required and the optional readings consist of technical readings and at least one practical application of the respective method.
- Students submit two **Homework assignments**, each of which counts towards 20% of the final grade. The assignments will be distributed as a [Quarto](#) file 14 days before the submission deadline. Students fill in the answers and solutions in the same Quarto file, rename it to `hw_01/02_surname_firstname.qmd`, render it as an `html` file, and submit it via Brightspace. Only rendered `html` files will be accepted! Homework 1 will be submitted at the end of Week 5. Homework 2 will be submitted at the end of Week 9. More details on the homework will be provided in the first session(s) of the course.
- Students also submit a short **Research Paper** of 3,000 words (excluding references and appendices). The research paper counts towards 60% of the final grade and must be submitted by 12 May 2025. In Homework 2, students will be required to briefly outline the research question they want to test in the research paper and describe which textual data they will use for testing this question.

In the research paper, the students should succinctly but clearly write up the results of a small research project using quantitative text analysis methods discussed in this module. Students are free to collect their own data or use existing data. Creativity is encouraged. Students are free to answer questions from all fields of political science or related disciplines but must justify their choice and the relevance of the question. This paper should contain the following elements:

Table 2: Grade conversion scheme for homeworks

Homeworks	UCD Grade
97–100%	A+
94–96%	A
91–93%	A–
88–90%	B+
85–87%	B
83–84%	B
80–82%	C+
77–79%	C
74–76%	C–
71–73%	D+
68–70%	D
65–67%	E+
54–64%	E
44–53%	E–
0–32%	F

1. Introduction and research question: introduction to the topic, research question, and relevance.
2. Expectations: a concise overview of the theoretical expectation(s) that will be tested in the results section.
3. Data and methods: description of the data sources as well as the methods employed.
4. Results: a discussion (with figures and tables) of the results of the analysis. This section forms the bulk of the paper.
5. Conclusion: a brief evaluation of the results and steps to push the research forward.

Overview of deadlines

Date	Assignment
End of Week 5	Homework 1 (20%)
End of Week 9	Homework 2 (20%)
Monday, 12 May 2025	Research Paper (60%)

Grading Criteria

In essence, markers assess four crucial elements in any answer:

- Analysis/understanding
- Extent and use of reading
- Organisation/structure
- Writing proficiency

The various grades/classifications listed below reflect the extent to which an answer displays essential features of each of these elements (and their relative weighting). At its simplest: the better the

analysis, the wider the range of appropriate sources consulted, the greater the understanding of the materials read, the clearer the writing style, and the more structured the argument, the higher will be the mark.

The following provides an indicative outline of the criteria used by markers to award a particular grade/classification. If you are in any confusion about how to correctly approach referencing and bibliography issues, please refer to the following guidelines: APSA Committee on Publications (2018). *Style Manual for Political Science (Revised 2018 Version)*. URL: <https://connect.apsanet.org/stylemanual/>.

Proper referencing is ESSENTIAL in a good assignment.

Grade Explanation for Research Paper

Grade: A (Excellent Performance)

A deep and systematic engagement with the assessment task, with consistently impressive demonstration of a comprehensive mastery of the subject matter, reflecting:

- A deep and broad knowledge and critical insight as well as extensive reading
- A critical and comprehensive appreciation of the relevant literature or theoretical, technical or professional framework
- An exceptional ability to organise, analyse and present arguments fluently and lucidly with a high level of critical analysis, amply supported by evidence, citation or quotation;
- A highly-developed capacity for original, creative and logical thinking
- An extensive and detailed knowledge of the subject matter
- A highly-developed ability to apply this knowledge to the task set
- Evidence of extensive background reading
- Clear, fluent, stimulating and original expression
- Excellent presentation (spelling, grammar, graphical) with minimal or no presentation errors
- Referencing style consistently executed in recognised style

Grade: B (Very Good Performance)

A thorough and well organised response to the assessment task, demonstrating:

- A thorough familiarity with the relevant literature or theoretical, technical or professional framework
- Well-developed capacity to analyse issues, organise material, present arguments clearly and cogently well supported by evidence, citation or quotation;
- Some original insights and capacity for creative and logical thinking
- A broad knowledge of the subject matter
- Considerable strength in applying that knowledge to the task set
- Evidence of substantial background reading
- Clear and fluent expression
- Quality presentation with few presentation errors
- Referencing style for the most part consistently executed in recognised style

Grade: C (Good Performance)

An intellectually competent and factually sound answer with, marked by:

- Evidence of a reasonable familiarity with the relevant literature or theoretical, technical or professional framework
- Good developed arguments, but more statements of ideas
- Arguments or statements adequately but not well supported by evidence, citation or quotation
- Some critical awareness and analytical qualities
- Some evidence of capacity for original and logical thinking
- Adequate but not complete knowledge of the subject matter
- Omission of some important subject matter or the appearance of several minor errors
- Capacity to apply knowledge appropriately to the task albeit with some errors
- Evidence of some background reading
- Clear expression with few areas of confusion
- Writing of sufficient quality to convey meaning but some lack of fluency and command of suitable vocabulary
- Good presentation with some presentation errors
- Referencing style executed in recognised style, but with some errors

Grade: D (Satisfactory Performance)

An acceptable level of intellectual engagement with the assessment task showing:

- Some familiarity with the relevant literature or theoretical, technical or professional framework
- Mostly statements of ideas, with limited development of argument
- Limited use of evidence, citation or quotation
- Limited critical awareness displayed
- Limited evidence of capacity for original and logical thinking
- Basic grasp of subject matter, but somewhat lacking in focus and structure
- Main points covered but insufficient detail
- Some effort to apply knowledge to the task but only a basic capacity or understanding displayed
- Little or no evidence of background reading
- Several minor errors or one major error
- Satisfactory presentation with an acceptable level of presentation errors
- Referencing style inconsistent

Grade: D– (Acceptable)

The minimum acceptable of intellectual engagement with the assessment task which:

- The minimum acceptable appreciation of the relevant literature or theoretical, technical or professional framework
- Ideas largely expressed as statements, with little or no developed or structured argument
- Minimum acceptable use of evidence, citation or quotation
- Little or no analysis or critical awareness displayed or is only partially successful
- Little or no demonstrated capacity for original and logical thinking
- Shows a basic grasp of subject matter but may be poorly focused or badly structured or contain irrelevant material
- Has one major error and some minor errors

- Demonstrates the capacity to complete only moderately difficult tasks related to the subject material
- No evidence of background reading
- Displays the minimum acceptable standard of presentation (spelling, grammar, graphical)
- Referencing inconsistent with major errors

Grade: E (Fail [marginal])

A factually sound answer with a partially successful, but not entirely acceptable, attempt to:

- Integrate factual knowledge into a broader literature or theoretical, technical or professional framework develop arguments
- Support ideas or arguments with evidence, citation or quotation
- Engages with the subject matter or problem set, despite major deficiencies in structure, relevance or focus
- Has two major error and some minor errors
- Demonstrates the capacity to complete only part of, or the simpler elements of, the task
- An incomplete or rushed answer (e.g. the use of bullet points through part/all of answer)
- Little or no referencing style evident

Grade: F (Fail [unacceptable])

An unacceptable level of intellectual engagement with the assessment task, with:

- No appreciation of the relevant literature or theoretical, technical or professional framework
- No developed or structured argument
- No use of evidence, citation or quotation
- No analysis or critical awareness displayed or is only partially successful
- No demonstrated capacity for original and logical thinking
- A failure to address the question resulting in a largely irrelevant answer or material of marginal relevance predominating
- A display of some knowledge of material relative to the question posed, but with very serious omissions / errors and/or major inaccuracies included in answer
- Solutions offered to a very limited portion of the problem set
- An answer unacceptably incomplete (e.g. for lack of time)
- A random and undisciplined development, layout or presentation
- Unacceptable standards of presentation, such as grammar, spelling or graphical presentation
- Evidence of substantial plagiarism
- No referencing style evident

Grade: G (Fail [wholly unacceptable])

No intellectual engagement with the assessment task

- Complete failure to address the question resulting in an entirely irrelevant answer
- Little or no knowledge displayed relative to the question posed
- Little or no solution offered for the problem set
- Evidence of extensive plagiarism
- No referencing style evident

Grade: NG (No Grade)

No work was submitted by the student or student was absent from the assessment, or work submitted did not merit a grade.

Course Structure

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Week 11: Fine-Tuning Transformer Models for Classification (17 April)	19
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Week 1: Introduction to Quantitative Text Analysis (23 January)

- What are quantitative text analysis and natural language processing?
- What is the structure of the module, and what are the expectations?
- *Application*: installing packages and setting up a project in VSCode

Mandatory Readings

- K. Benoit (2020). “Text as Data: An Overview”. *Handbook of Research Methods in Political Science and International Relations*. Ed. by L. Curini and R. Franzese. Thousand Oaks: Sage: 461–497.

- J. Grimmer and B. M. Stewart (2013). “Text as Data: The Promise and Pitfalls of Automatic Content Analysis Methods for Political Texts”. *Political Analysis* 21 (3): 267–297.

Optional

- K. L. Nielbo, F. Karsdorp, M. Wevers, A. Lassche, B. R. B., M. Kestemont, and N. Tahmasebi (2024). “Quantitative Text Analysis”. *Nature Reviews Methods Primers* 2 (24)
- J. Wilkerson and A. Casas (2017). “Large-Scale Computerized Text Analysis in Political Science: Opportunities and Challenges”. *Annual Review of Political Science* 20: 529–544.
- M. Gentzkow, B. T. Kelly, and M. Taddy (2019). “Text as Data”. *Journal of Economic Literature* 57 (3): 535–574.

Week 2: Workflow, R and Quarto (30 January)

- What are the underlying assumptions of text-as-data approaches?
- How to set up and use R?
- *Application*: overview of important R functions; structure the workflow for a quantitative research project?

Mandatory Readings

- K. Watanabe and S. Müller (2023). *Quanteda Tutorials*. URL: <https://tutorials.quanteda.io>: chapter 1.
- J. Grimmer, M. E. Roberts, and B. M. Stewart (2022). *Text as Data: A New Framework for Machine Learning and the Social Sciences*. Princeton: Princeton University Press: chapter 4.
- R. Alexander (2023). *Telling Stories with Data: With Applications in R*. New York: CRC Press: chapter 3.

Optional

- E. Ash and S. Hansen (2023). “Text Algorithms in Economics”. *Annual Review of Economics* 15 (659-688).
- K. Benoit, K. Watanabe, H. Wang, P. Nulty, A. Obeng, S. Müller, and A. Matsuo (2018). “quanteda: An R Package for the Quantitative Analysis of Textual Data”. *The Journal of Open Source Software* 3 (30): 774.
- C. D. Manning, P. Raghavan, and H. Schütze (2008). *An Introduction to Information Retrieval*. New York: Cambridge University Press: chapter 2.

Week 3: Tokenisation and Document-Feature Matrix (6 February)

- What are tokens, types, and features?
- What is the difference between stemming and lemmatisation?
- What information can we extract from a document-feature matrix?
- *Application*: tokenising texts, removing features, and creating a document-feature matrix

Mandatory Readings

- J. Grimmer, M. E. Roberts, and B. M. Stewart (2022). *Text as Data: A New Framework for Machine Learning and the Social Sciences*. Princeton: Princeton University Press: chapter 5.
- K. Watanabe and S. Müller (2023). *Quanteda Tutorials*. URL: <https://tutorials.quanteda.io>: chapter 2–3.

Optional

- M. W. Denny and A. Spirling (2018). “Text Preprocessing For Unsupervised Learning: Why It Matters, When It Misleads, And What To Do About It”. *Political Analysis* 26 (2): 168–189.
- K. Welbers, W. Van Atteveldt, and K. Benoit (2017). “Text Analysis in R”. *Communication Methods and Measures* 11 (4): 245–265.

Week 4: Dictionaries and Sentiment: Old and New Approaches (13 February)

- What are automated dictionary approaches? How can we create, validate, refine, and apply dictionaries? Can (or should) LLMs replace dictionaries?
- *Application*: creating multi-word expressions and applying dictionaries to tokens objects and document-feature matrices

Mandatory Readings

- S.-O. Proksch, W. Lowe, J. Wäckerle, and S. N. Soroka (2019). “Multilingual Sentiment Analysis: A New Approach to Measuring Conflict in Legislative Speeches”. *Legislative Studies Quarterly* 44 (1): 97–131.
- S. Müller (2020). “Media Coverage of Campaign Promises Throughout the Electoral Cycle”. *Political Communication* 37 (5): 696–718.
- S. Rathje, D.-M. Mirea, I. Sucholutsky, R. Marjeh, C. E. Robertson, and J. J. Van Bavel (2024). “GPT is an Effective Tool for Multilingual Psychological Text Analysis”. *Proceedings of the National Academy of Sciences of the United States of America (PNAS)* 121 (34): e2308950121.

Optional

- A. Muddiman, S. C. McGregor, and N. J. Stroud (2019). “(Re)Claiming Our Expertise: Parsing Large Text Corpora With Manually Validated and Organic Dictionaries”. *Political Communication* 36 (2): 214–226.
- D. Jurafsky and J. H. Martin (2024). *Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition with Language Models*. 3rd edition: ch. 22.

Week 5: Describing and Comparing Texts (20 February)

- How do texts differ in their ‘readability’ and complexity? What are measures to estimate the similarity and distance between texts?

- How can we identify distinct features in texts?
- What are word embeddings and how can we use them for social science research?
- *Application*: estimating readability, similarity, and “keyness”; train Word2Vec model and identify similarities between word vectors

Mandatory Readings

- K. Benoit, K. Munger, and A. Spirling (2019). “Measuring and Explaining Political Sophistication Through Textual Complexity”. *American Journal of Political Science* 63 (2): 491–508.
- P. L. Rodriguez and A. Spirling (2022). “Word Embeddings: What Works, What Doesn’t, and How to Tell the Difference for Applied Research”. *The Journal of Politics* 84 (1): 101–115.

Optional

- E. Hengel (2022). “Publishing While Female: Are Women Held to Higher Standards? Evidence from Peer Review”. *The Economic Journal* 132 (648): 2951–2991.
- H. C. Shulman, D. M. Markowitz, and T. Rogers (2024). “Reading Dies in Complexity: Online News Consumers Prefer Simple Writing”. *Science Advances* 10 (23): eadn2555.
- D. Bischof and R. Senninger (2018). “Simple Politics for the People? Complexity in Campaign Messages and Political Knowledge”. *European Journal of Political Research* 57 (2): 473–495.
- E. M. Wirsching, P. L. Rodriguez, A. Spirling, and B. M. Stewart (2024). “Multilanguage Word Embeddings for Social Scientists: Estimation, Inference, and Validation Resources for 157 Languages”. *Political Analysis* published ahead of print (doi: 10.1017/pan.2024.17).
- M. R. Holman, R. Johnson, and T. Simko (2024). *Measuring Conflict in Local Politics*. OSF PrePrint. URL: <https://osf.io/vst9g>.
- J. Blumenau (2021). “The Effects of Female Leadership on Women’s Voice in Political Debate”. *British Journal of Political Science* 51 (2): 750–771.

Week 6: Human Coding and Document Classification (27 February)

- How can we classify documents into known and pre-defined categories? How do we create a training set? How do we assess the classification performance?
- Difference between bag-of-words classifier and state-of-the art approaches (transformers)
- *Application*: supervised machine learning in `quanteda` and the `transformers` Python library

Mandatory Readings

- S. Müller (2022). “The Temporal Focus of Campaign Communication”. *The Journal of Politics* 84 (1): 585–590.
- S. Lacy, B. R. Watson, D. Riffe, and J. Lovejoy (2015). “Issues and Best Practices in Content Analysis”. *Journalism & Mass Communication Quarterly* 92 (4): 791–811.

- F. Gilardi, M. Alizadeh, and M. Kubli (2023). “ChatGPT Outperforms Crowd-Workers for Text-Annotation Tasks”. *Proceedings of the National Academy of Sciences of the United States of America* 120 (3): e2305016120.

Optional

- C. Hanretty (2023). *Replicating Mueller, ‘The Temporal Focus of Campaign Communication’*. URL: <https://www.chrishanretty.co.uk/posts/finetuning/>.
- S. Kapoor and A. Narayanan (2023). “Leakage and the Reproducibility Crisis in Machine-Learning-Based Science”. *Patterns* 4: 100804.
- A. Peterson and A. Spirling (2018). “Classification Accuracy as a Substantive Quantity of Interest: Measuring Polarization in Westminster Systems”. *Political Analysis* 26 (1): 120–128.
- B. Castanho Silva, D. Pullan, and J. Wäckerle (2024). “Blending In or Standing Out? Gendered Political Communication in 24 Democracies”. *American Journal of Political Science* published ahead of print (doi: 10.1111/ajps.12876).
- K. Benoit, D. Conway, B. E. Lauderdale, M. Laver, and S. Mikhaylov (2016). “Crowd-Sourced Text Analysis: Reproducible and Agile Production of Political Data”. *American Political Science Review* 110 (2): 278–295.
- L. Birkenmaier, C. M. Lechner, and C. Wagner (2024). “The Search for Solid Ground in Text as Data: A Systematic Review of Validation Practices and Practical Recommendations for Validation”. *Communication Methods and Measures* 18 (3): 249–277.

Week 7: Supervised, Unsupervised and Semi-Supervised Scaling (6 March)

- What are the assumptions, advantages, and problems of supervised and unsupervised scaling?
- How can we use supervised scaling to answer substantive questions?
- *Application*: Wordscores, Wordfish, and Latent Semantic Scaling

Mandatory Readings

- M. Laver, J. Garry, and K. Benoit (2003). “Extracting Policy Positions from Political Texts Using Words as Data”. *American Political Science Review* 97 (2): 311–331.
- J. B. Slapin and S.-O. Proksch (2008). “A Scaling Model for Estimating Time-Series Party Positions from Texts”. *American Journal of Political Science* 52 (3): 705–722.
- K. Watanabe (2021). “Latent Semantic Scaling: A Semisupervised Text Analysis Technique for New Domains and Languages”. *Communication Methods and Measures* 14 (2): 81–102.

Optional

- G. Le Mens and A. Gallego (2024). *Positioning Political Texts with Large Language Models by Asking and Averaging*. arXiv:2311.16639. URL: <https://doi.org/10.48550/arXiv.2311.16639>.
- D. Zollinger (2024). “Cleavage Identities in Voters’ Own Words: Harnessing Open-Ended Survey Responses”. *American Journal of Political Science* 68 (1): 139–159.

- S. Müller and N. Fujimura (2024). “Campaign Communication and Legislative Leadership”. *Political Science Research and Methods* published ahead of print (doi: 10.1017/psrm.2024.11).
- T. Gessler and S. Hunger (2022). “How the Refugee Crisis and Radical Right Parties Shape Party Competition on Immigration”. *Political Science Research and Methods* 10 (3): 524–544.
- T. O’Grady (2019). “Careerists Versus Coal-Miners: Welfare Reforms and the Substantive Representation of Social Groups in the British Labour Party”. *Comparative Political Studies* 52 (4): 544–578.
- S. Müller, S. Brazys, and A. Dukalskis (2024). “Discourse Wars and ‘Mask Diplomacy’: China’s Global Image Management in Times of Crisis”. *Political Research Exchange* 6 (1): 2337632.

Week 8: Retrieving, Loading and Wrangling Text Corpora (27 March)

- What are typical text corpora you can use for your final research paper?
- What are APIs and how can we use them to retrieve data?
- How can we load various types of text corpora and transform them into a quantified corpus object?
- What are legal and ethical requirements and challenges when working with social media data?
- *Application*: Manifesto Corpus, UN General Debate Corpus, Guardian API, Twitter API

Mandatory Readings

- H. Wickham, M. Çetinkaya-Runde, and G. Grolemund (2023). *R for Data Science: Import, Tidy, Transform, Visualize, and Model Data*. 2nd edition. Sebastopol: O’Reilly: ch. 5–7.
- P. C. Bauer and C. Landesvatter, eds. (2023). *APIs for Social Scientists: A Collaborative Review*: skim potentially relevant chapters.

Optional

- N. Merz, S. Regel, and J. Lewandowski (2016). “The Manifesto Corpus: A New Resource for Research on Political Parties and Quantitative Text Analysis”. *Research & Politics* 3 (2): 1–8.
- S. Barari and T. Simko (2023). “LocalView, a Database of Public Meetings for the Study of Local Politics and Policy-making in the United States”. *Scientific Data* 10: 135.
- M. J. Salganik (2017). *Bit by Bit: Social Research in the Digital Age*. Princeton: Princeton University Press: ch. 6.
- A. Fourney (2024). *MarkItDown: Python Tool for Converting Files and Office Documents to Markdown*. Version 0.0.1a3. URL: <https://pypi.org/project/markitdown/>.

Week 9: Topic Models (3 April)

- How does unsupervised document classification work? What are the assumptions, advantages, and caveats of topic models?
- *Application*: Compare unseeded and keyword-assisted topic models

Mandatory Readings

- T. Gessler (2022). “[Topic Models](#)”. *Elgar Encyclopedia of Technology and Politics*. Ed. by A. Ceron. Cheltenham: Edward Elgar Publishing: 108–111.
- S. Eshima, K. Imai, and T. Sasaki (2024). “[Keyword-Assisted Topic Models](#)”. *American Journal of Political Science* 68 (2): 730–750.

Optional

- M. E. Roberts, B. M. Stewart, D. Tingley, C. Lucas, J. Leder-Luis, S. K. Gadarian, B. Albertson, and D. G. Rand (2014). “[Structural Topic Models for Open-Ended Survey Responses](#)”. *American Journal of Political Science* 58 (4): 1064–1082.
- R. Parthasarathy, V. Rao, and N. Palaniswamy (2019). “[Deliberative Democracy in an Unequal World: A Text-As-Data Study of South India’s Village Assemblies](#)”. *American Political Science Review* 113 (3): 623–640.
- S. Müller, G. Kennedy, and T. Maher (2023). “[Reactions to Experts in Deliberative Democracy: The 2016–2018 Irish Citizens’ Assembly](#)”. *Irish Political Studies* 38 (4): 467–488.
- A. Catalinac (2016). “[From Pork to Policy: The Rise of Programmatic Campaigning in Japanese Elections](#)”. *The Journal of Politics* 78 (1): 1–18.

Week 10: Transformers: Fundamentals (10 April)

- Basic knowledge of the inner workings of transformers
- *Application*: Exploring the `transformers` Python library and HuggingFace infrastructure; using existing models for classification

Mandatory Readings

- A. Kroon, K. Welbers, D. Trilling, and W. van Atteveldt (2024). “[Advancing Automated Content Analysis for a New Era of Media Effects Research: The Key Role of Transfer Learning](#)”. *Communication Methods and Measures* 18 (2): 142–162.
- Financial Times Visual Storytelling Team (2023). *Generative AI Exists Because of the Transformer*. URL: <https://ig.ft.com/generative-ai/>.

Optional

- D. Jurafsky and J. H. Martin (2024). *Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition with Language Models*. 3rd edition: ch. 9 (The Transformer).
- S. Wankmüller (2024). “[Introduction to Neural Transfer Learning with Transformers for Social Science Text Analysis](#)”. *Sociological Methods & Research* 53 (4): 1676–1752.
- L. Tunstall, L. von Werra, and T. Wolf (2022). *Natural Language Processing with Transformers: Building Language Applications with Hugging Face*. Beijing: O’Reilly.

Week 11: Fine-Tuning Transformer Models for Classification (17 April)

- Fine-tune a pre-trained transformer model for a domain-specific classification task
- *Application*: Using the `transformers` Python library to fine-tune transformer for domain-specific task

Mandatory Readings

- HuggingFace (2023). *Transformers: State-of-the-art Machine Learning for PyTorch, TensorFlow, and JAX*. V4.35.0/V4.35.1/V4.35.2. URL: https://huggingface.co/docs/transformers/skim_tutorials for a basic understanding of the transformers library.
- M. J. J. Bucher and M. Martini (2024). *Fine-Tuned ‘Small’ LLMs (Still) Significantly Outperform Zero-Shot Generative AI Models in Text Classification*. arXiv PrePrint. URL: <https://arxiv.org/abs/2406.08660>.

Optional

- S. Müller and S.-O. Proksch (2024). “Nostalgia in European Party Politics: A Text-Based Measurement Approach”. *British Journal of Political Science* 54 (3): 993–1005.
- G. Okasa, A. de León, M. Strinzel, A. Jorstad, K. Milzow, M. Egger, and S. Müller (2024). *A Supervised Machine Learning Approach for Assessing Grant Peer Review Reports*. arXiv PrePrint. URL: <https://arxiv.org/abs/2411.16662>.
- B. Warner, A. Chaffin, B. Clavié, O. Weller, O. Hallström, S. Taghadouini, A. Gallagher, R. Biswas, F. Ladhak, T. Aarsen, N. Cooper, G. Adams, J. Howard, and I. Poli (2024). *Smarter, Better, Faster, Longer: A Modern Bidirectional Encoder for Fast, Memory Efficient, and Long Context Finetuning and Inference*. arXiv PrePrint. URL: <https://arxiv.org/abs/2412.13663>.
- Z. P. Dickson and S. B. Hobolt (2024). “Going Against the Grain: Climate Change as a Wedge Issue for the Radical Right”. *Comparative Political Studies* published ahead of print (doi: 10.1177/0010414024127129).

Week 12: Large Language Models (24 April)

- Using Large Language Models (LLMs) for text classification and summarisation
- Differences between zero-shot and few-shot approaches
- Challenges: replicability and open science principles
- *Application*: Running LLMs, such as GPT and LLama, through APIs and locally

Mandatory Readings

- J. Cova and L. Schmitz (2024). *A Primer for the Use of Classifier and Generative Large Language Models in Social Science Research*. OSF PrePrint. URL: <https://doi.org/10.31219/osf.io/r3qng>.
- J. T. Ornstein, E. N. Blasingame, and J. S. Truscott (2025). “How to Train Your Stochastic Parrot: Large Language Models for Political Texts”. *Political Science Research and Methods* published ahead of print (doi: 10.1017/psrm.2024.64).

- C. Barrie, A. Palmer, and A. Spirling (2024). *Replication for Language Models: Problems, Principles, and Best Practices for Political Science*. URL: https://arthurspirling.org/documents/BarriePalmerSpirling_TrustMeBro.pdf.

Optional

- A. Spirling (2023). “Open Generative AI Models are a Way Forward for Science”. *Nature* 616: 413.
- T. Hu, Y. Kyrychenko, S. Rathje, N. Collier, S. van der Linden, and J. Roozenbeek (2025). “Generative Language Models Exhibit Social Identity Biases”. *Nature Computational Science* 5: 65–75.
- D. Jurafsky and J. H. Martin (2024). *Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition with Language Models*. 3rd edition: ch. 10 (Large Language Models).
- I. Haaland, C. Roth, S. Stantcheva, and J. Wohlfart (2024). “Understanding Economic Behavior Using Open-ended Survey Data”. NBER Working Paper 32421.

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- Barari, S. and T. Simko (2023). “LocalView, a Database of Public Meetings for the Study of Local Politics and Policy-making in the United States”. *Scientific Data* 10: 135.
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- Benoit, K., D. Conway, B. E. Lauderdale, M. Laver, and S. Mikhaylov (2016). “Crowd-Sourced Text Analysis: Reproducible and Agile Production of Political Data”. *American Political Science Review* 110 (2): 278–295.
- Benoit, K., K. Munger, and A. Spirling (2019). “Measuring and Explaining Political Sophistication Through Textual Complexity”. *American Journal of Political Science* 63 (2): 491–508.
- Benoit, K., K. Watanabe, H. Wang, P. Nulty, A. Obeng, S. Müller, and A. Matsuo (2018). “quanteda: An R Package for the Quantitative Analysis of Textual Data”. *The Journal of Open Source Software* 3 (30): 774.
- Birkenmaier, L., C. M. Lechner, and C. Wagner (2024). “The Search for Solid Ground in Text as Data: A Systematic Review of Validation Practices and Practical Recommendations for Validation”. *Communication Methods and Measures* 18 (3): 249–277.
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- Denny, M. W. and A. Spirling (2018). “Text Preprocessing For Unsupervised Learning: Why It Matters, When It Misleads, And What To Do About It”. *Political Analysis* 26 (2): 168–189.
- Dickson, Z. P. and S. B. Hobolt (2024). “Going Against the Grain: Climate Change as a Wedge Issue for the Radical Right”. *Comparative Political Studies* published ahead of print (doi: 10.1177/0010414024127129).
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- Grimmer, J., M. E. Roberts, and B. M. Stewart (2022). *Text as Data: A New Framework for Machine Learning and the Social Sciences*. Princeton: Princeton University Press.
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- Haaland, I., C. Roth, S. Stantcheva, and J. Wohlfart (2024). “Understanding Economic Behavior Using Open-ended Survey Data”. NBER Working Paper 32421.
- Hanretty, C. (2023). *Replicating Mueller, ‘The Temporal Focus of Campaign Communication’*. URL: <https://www.chrishanretty.co.uk/posts/finetuning/>.
- Healy, K. (2019). *Data Visualization: A Practical Introduction*. Princeton: Princeton University Press.
- Hengel, E. (2022). “Publishing While Female: Are Women Held to Higher Standards? Evidence from Peer Review”. *The Economic Journal* 132 (648): 2951–2991.
- Holman, M. R., R. Johnson, and T. Simko (2024). *Measuring Conflict in Local Politics*. OSF PrePrint. URL: <https://osf.io/vst9g>.
- Hu, T., Y. Kyrchenko, S. Rathje, N. Collier, S. van der Linden, and J. Roozenbeek (2025). “Generative Language Models Exhibit Social Identity Biases”. *Nature Computational Science* 5: 65–75.
- HuggingFace (2023). *Transformers: State-of-the-art Machine Learning for PyTorch, TensorFlow, and JAX*. V4.35.0/V4.35.1/V4.35.2. URL: <https://huggingface.co/docs/transformers/>.
- Hvitfeldt, E. and J. Silge (2021). *Supervised Machine Learning For Text Analysis in R*. Boca Raton: CRC Press.
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- Jurafsky, D. and J. H. Martin (2024). *Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition with Language Models*. 3rd edition.
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- Muddiman, A., S. C. McGregor, and N. J. Stroud (2019). “(Re)Claiming Our Expertise: Parsing Large Text Corpora With Manually Validated and Organic Dictionaries”. *Political Communication* 36 (2): 214–226.
- Müller, S. (2020). “Media Coverage of Campaign Promises Throughout the Electoral Cycle”. *Political Communication* 37 (5): 696–718.
- Müller, S. (2022). “The Temporal Focus of Campaign Communication”. *The Journal of Politics* 84 (1): 585–590.
- Müller, S. (2023). “How Slack Facilitates Communication and Collaboration in Seminars and Project-Based Courses”. *Journal of Educational Technology Systems* 51 (3): 303–316.
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